A Primer in Discrete Probability

Dominik Scheder

March 27, 2019

Abstract

1 Finite Probability Spaces

Here is a random experiment all of us have done in our lives: rolling a die. There are six possible outcomes, 1,2,3,4,5, and 6. Assuming that the die is

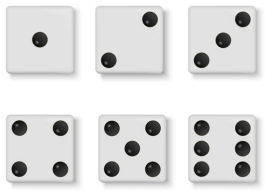


Figure 1: The six sides of a die.

"fair", each outcome is equally likely, and thus has probability 1/6. This is a toy example of a *probability space*.

Definition 1. *A* finite probability space *is a finite set S together with a function $P : S \to \mathbb{R}$ such that $P(s) \geq 0$ for every $s \in S$ and $\sum_{s \in S} P(s) = 1$.*

In our die-rolling example, S would be the set $\{1, 2, 3, 4, 5, 6\}$ and P is the function that maps every element to 1/6. Suppose we roll two dice, a white and a red one. We get another probability space. This probability space has

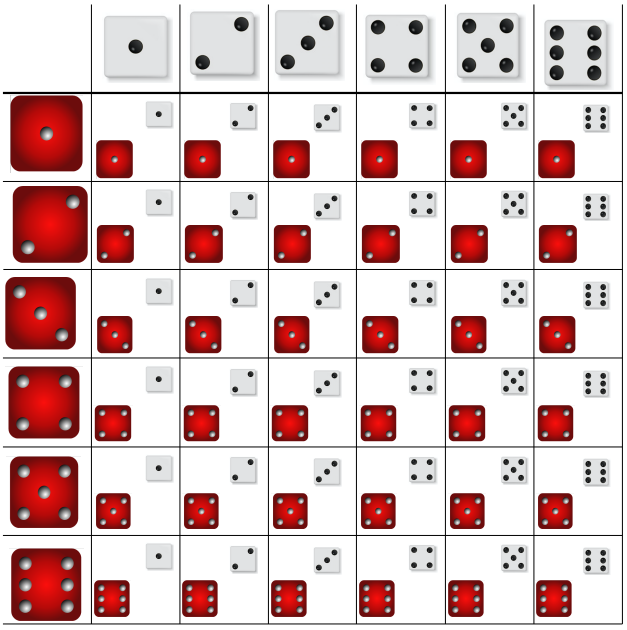


Figure 2: Rolling two dice.

36 elements, all possible combinations of rolling two dice. We could write down S explicitly, as $S = \{(1,1),(1,2),(1,3),\ldots,(6,4),(6,5),(6,6)\}$, if we wanted to.

1.1 Events

Sometimes, at least one of the two rolled dice shows a 5 or higher. Sometimes not, and then both dice are 4 or lower. What is the probability of this happening? Let $\mathcal{E}$ denote the event that at least one of the two dice is 5 or higher. What kind of object is $\mathcal{E}$, mathematically speaking? Formally, $\mathcal{E}$ is a subset of the probability space: $\mathcal{E} \subseteq S$. Indeed,

$$\begin{aligned}\mathcal{E} = \{&(1,5),(1,6),(2,5),(2,6),(3,5),(3,6),(4,5),(4,6),\\ &(5,1),(5,2),(5,3),(5,4),(5,5),(5,6),\\ &(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\}\ .\end{aligned}$$

So what is the probability of $\mathcal{E}$? There are 36 possible outcomes, i.e., $|S| = 36$; for 20 of them $\mathcal{E}$ happens, i.e., $|\mathcal{E}| = 20$. The total probability of all those

elements in $\mathcal{E}$ is

$$P(\mathcal{E}) = \sum_{s \in S} P(s) = \frac{|\mathcal{E}|}{|S|} = \frac{20}{36} = \frac{5}{9} \ .$$

Definition 2. *Let S be a finite probability space. An event is a subset $\mathcal{E} \subseteq S$ and its probability is $P(\mathcal{E}) := \sum_{s \in S} P(s)$.*

1.2 Independence

Let us come up with a different, more elegant way of computing $P(\mathcal{E})$. We define two events A and B as follows: A is the event that the first die is 4 or lower; B is the event that the second die is 4 or lower. Let us observe three simple facts:

1. $P(A) = 4/6$.
2. $P(B) = 4/6$.
3. $\mathcal{E} = S \setminus (A \cap B)$.

How does this help us to compute $P(\mathcal{E})$? We observe that A and B are independent events.

Definition 3. *Let $A, B \subseteq S$ be two events. We say A and B are independent if $P(A \cap B) = P(A) \cdot P(B)$.*

You are encouraged to check that this indeed holds for our two events A and B. Intuitively, A talks only about the first die, and B only about the second, and the two rolls are independent. We can compute:

$$P(\mathcal{E}) = 1 - P(A \cap B) = 1 - P(A) \cdot P(B) = 1 - \frac{4}{6} \cdot \frac{4}{6} = \frac{5}{9} \ .$$

Suppose we roll a die n times. Let $\mathcal{E}$ be the event that we see at least one 6. What is $P(\mathcal{E})$? For $i = 1, \ldots, n$, let A_i denote the event that the ith die is 5 or lower. Clearly, $P(A_i) = 5/6$ and $\mathcal{E} = S \setminus (A_1 \cap \cdots \cap A_n)$. Observe that each A_i "talks about a different die", so they are all independent.

Definition 4. *Let $A_1, \ldots, A_n \subseteq S$ be events in a probability space S. We say $A_1, \ldots, A_n$ are* independent *if*

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i) \tag{1}$$

holds for all sets $I \subseteq [n]$.

Intuitively, since A_i talks about a different die, they should be independent. You are invited to verify this more formally. We compute

$$P(\mathcal{E}) = 1 - P(A_1 \cap \cdots \cap A_n) = 1 - \left(\frac{5}{6}\right)^n \ .$$

Note: Often we write $\Pr[\mathcal{E}]$ instead of $P(\mathcal{E})$, if P is clear from the context. The reason is that the reader will immediately recognize $\Pr[\cdot]$ as a probability, whereas $P(\cdot)$ can have many meanings in mathematics.

1.3 Example: Bit Strings Without 11

Consider the set

$$A_n := \{x \in \{0,1\}^n \mid x \text{ does not contain the substring } 11\}$$

We have seen in the lecture that $|A_n| = F_{n+2}$, the $(n+2)^{\text{nd}}$ Fibonacci number. Let P_n be the uniform distribution over A_n. That is, $P_n(x) = \frac{1}{|A_n|}$ for every $x \in A_n$. So (A_n, P_n) form a probability space.

Exercise 1.1. Let Q_n^i be the event that $x_i = 1$.

1. Give an explicit formula for $\Pr[Q_n^i]$. Recall the explicit formula for the Fibonacci numbers.

2. Give a simpler, "asymptotic" formula for $\Pr[Q_n^i]$, if n is large and i is neither close to 0 nor to 1. For example, take some $0 < \delta < 1$ and compute the limit

   $$q := \lim_{n\to\infty} \Pr[Q_n^i] \ , \text{ where } i = \lfloor \delta n \rfloor$$

   This is to say, "each bit of x not too close to the left or right is 1 with probability close to q".

3. Show that the events Q_n^i and Q_n^j are not independent.

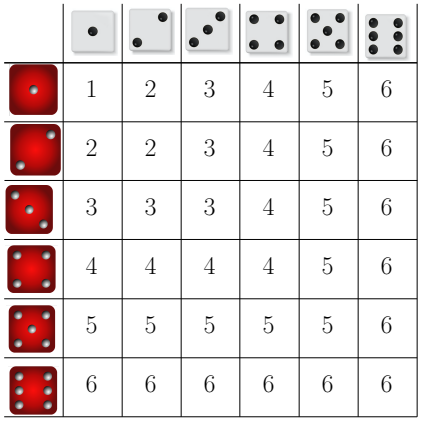
4. Show that they are "asymptotically independent" if 1, i, j, and n are sufficiently far apart. More precisely, show that for any $0 < \alpha < \beta < 1$, it holds that

   $$\lim_{n\to\infty} \Pr[Q_n^i \cap Q_n^j] = \lim_{n\to\infty} \Pr[Q_n^i] \cdot \Pr[Q_n^j] \ ,$$

   where $i = \lfloor \alpha n \rfloor$ and $j = \lfloor \beta n \rfloor$.

2 Random Variables

Assume again that we roll two dice, and let X denote the value of the higher one. Obviously, X can take on values $1, 2, 3, 4, 5, 6$, but higher values will be more likely. Formally, a random variable is a function $S \to \mathbb{R}$. In this



	1	2	3	4	5	6
	2	2	3	4	5	6
	3	3	3	4	5	6
	4	4	4	4	5	6
	5	5	5	5	5	6
	6	6	6	6	6	6

Figure 3: The maximum value of two dice.

concrete example, an element of S has the form (s_1, s_2), and the random variable X is the function

$$X : S \to \mathbb{R}, \quad (a_1, a_2) \mapsto \max(a_1, a_2) \ .$$

What are random variables good for? First of all, they simplify notation. Our first event $\mathcal{E}$ that at least one die has value 5 or larger, can be conveniently written as "$X \geq 5$". Its complement, that both are 4 or smaller, as "$X \leq 4$". Thus,

$$P(X \geq 5) = \frac{5}{9} \ .$$

Second, we can talk about "average outcomes".

2.1 Expectation

What is the "average value" of X? The possible outcomes of X are $1, 2, 3, 4, 5, 6$, but larger values are more likely. In particular,

i	1	2	3	4	5	6
$P(X = i)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

Thus, the average value of X is

$$\frac{1}{36} \times 1 + \frac{3}{36} \times 2 + \frac{5}{36} \times 3 + \frac{7}{36} \times 4 + \frac{9}{36} \times 5 + \frac{11}{36} \times 6 = \frac{161}{36} = 4.7222... \ .$$

Definition 5. *Let (S, P) be a finite probability space and $X : S \to \mathbb{R}$ a random variable. The* expectation of S*, denoted $\mathbb{E}[X]$, is*

$$\mathbb{E}[X] := \sum_{s \in S} P(s)X(s) \ .$$

Let $X, Y : S \to \mathbb{R}$ be two random variables. Then $X + Y$ a random variable, too.

Lemma 6 (Linearity of Expectation)**.** $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$.

Proof. Let us write down the definition of $\mathbb{E}[X+Y]$ and perform some simple manipulations:

$$\begin{aligned}
\mathbb{E}[X + Y] &= \sum_{s \in S} P(s) \left(X(s) + Y(s)\right) \\
&= \sum_{s \in S} \left(P(s)X(s) + P(s)Y(s)\right) \\
&= \sum_{s \in S} P(s)X(s) + \sum_{s \in S} P(s)Y(s) \\
&= \mathbb{E}[X] + \mathbb{E}[Y] \ .
\end{aligned}$$

□

Note that the proof only uses the laws of associativity and commutativity. It is not required that X, Y be independent. Wait: what does it mean that "X and Y are independent"? We have defined independence only for *events*, not *random variables*. Well, let's fix that.

Definition 7. *Two random variables X, Y are* independent *if for all $a, b \in S$, the two events $[X = a]$ and $[Y = b]$ are independent. That is, if $P(X = a \wedge Y = b) = P(X = a) \cdot P(Y = b)$. In general, n random variables $X_1, \ldots, X_n$ are independent if for all $a_1, \ldots, a_n \in S$, the events $[X_1 = a_1], \ldots, [X_n = a_n]$ are independent.*

Exercise 2.1. Show that $\mathbb{E}[X \cdot Y] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$ if X, Y are independent.

2.2 Using Random Variables

Let us a coin. For head, let's write a 1, for tail, a 0. Let's do this n times, which produces a string of length n. This is our probability space:

$$S = \{0,1\}^n \ .$$

Every string has the same probability 2^{-n}. Let X be the number of times "11" appears in the string. For example, in 11011110 it appears 4 times, starting at positions 1, 4, 5, 6. What is $\mathbb{E}[X]$?

s	$X(s)$
000	0
001	0
010	0
011	1
100	0
101	0
110	1
111	2

Alright, so for $n = 3$ we get $\mathbb{E}[X] = \frac{1}{8}(1 + 1 + 2) = \frac{1}{2}$. Can we get an exact formula for general n? Think about the following: what is $\Pr[X = j]$? In other words, how many bit strings of length n have exactly j occurrences of 11? Try to find a formula for this. I think it will be extremely messy. Fortunately, there is a powerful tool called *indicator variables*. Define $X_1, \ldots, X_{n-1}$ as follows: X_i is 1 if $s_i = s_{i+1} = 1$; that is, if the pattern 11 occurs in s at position i. Note that X_i is a random variable. We often use the notation

$$X_i := [s_i = s_{i+1} = 1] \ .$$

The expression "[statement]" evaluates to 1 if "statement" is true and to 0 otherwise. This is an extremely convenient notation yet strangely not completely standard in mathematics. With this notation we observe that $X = X_1 + X_2 + \cdots + X_{n_1}$. Indeed, every X_i indicates whether 11 occurs at position i (thus *indicator variable*), so their sum equals the number of occurrences. We compute the expectation of X_i:

$$\mathbb{E}[X_i] = 1 \cdot \Pr[X_i = 1] + 0 \cdot \Pr[X_i = 0] = \Pr[X_i = 1] = \Pr[s_i = s_{i+1} = 1] = \frac{1}{4} \ .$$

Note that X_i, X_{i+1} are *not* independent. Still, using linearity of expectation we compute

$$\mathbb{E}[X] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_{n-1}] = \frac{n-1}{4} \ .$$

3 Tossing a Coin Infinitely Often

Imagine we toss a fair coin until we see three heads in a row. We kind of have an intuitive understanding of the randomness involved, but what is the underlying probability space, formally? Well, there is no upper bound on how often we have to toss, so our probability space is

$$S := \{0,1\}^\infty = \{(x_1x_2x_3\dots), x_i \in \{0,1\}\ \}\ ,$$

the set of all infinite bit sequences. This is a new situation: an infinite probability space, actually *uncountably infinite*. How should we define the probability function $P : S \to \mathbb{R}$? For example, what is $P(000\dots)$, the probability we toss only zeroes? One thing should be clear: if we only talk about the first n tosses, they should just be uniformly distributed over $\{0,1\}^n$. So for example it should hold that

$$\Pr[\text{ the first } n \text{ tosses are } 0] = 2^{-n}\ .$$

So clearly $P(000\dots) \le 2^{-n}$. Indeed, tossing only zeroes means in particular that the first n tosses are 0. Note that $P(000\dots) \le 2^{-n}$ holds for every number $n \in \mathbb{N}$. Thus, we can only conclude that $P(000\dots) = 0$. The same argument works for any other infinite sequence: $P(x) = 0$ for every $x \in S$. Now this seems counter-intuitive. If we toss the coin infinitely many times, *something* must come up, but every concrete string has probability 0.

The way out of this uncomfortable situation requires a bit of formal machinery, which we do not want to introduce now. Let us stick to the following observation: events that only talk about the first n bits have a well-defined probability, since we can fall back on the discrete space $\{0,1\}^n$ with the uniform distribution. For example,

$$\Pr[\text{ the first } n \text{ tosses do not contain the pattern } 11] = \frac{|A_n|}{2^n} = \frac{F_{n+2}}{2^n}\ .$$

How about the event $\mathcal{E}$ that 11 appears earlier than 10? Certainly, this event does not only "talk about the first n bits"—there is no upper bound on how often we have to toss the coin until 11 or 10 appears. There is a nice way out: define $\mathcal{E}_n$ to be the event that 11 appears first after n coin tosses but 10 does not appear in $x_1x_2\dots x_n$. A moment of thought shows that

$$\mathcal{E}_n = [x_1 = x_2 = \dots = x_{n-2} = 0, x_{n-1} = x_n = 1]\ .$$

So obviously $\mathcal{E}_n$ talks only about the first n bits and thus $P(\mathcal{E}_n)$ is defined; in this case, it is 2^{-n}. Now $\mathcal{E}$ happens if and only if one of $\mathcal{E}_n$ happens:

$$\mathcal{E} = \bigcup_{n \geq 2} \mathcal{E}_n \ ,$$

and none of the events $\mathcal{E}_2, \mathcal{E}_3, \ldots$ can happen simultaneously. Therefore it is reasonable that

$$\Pr[\mathcal{E}] = \sum_{n \geq 2} \Pr[\mathcal{E}_n] \ .$$

If you continue thinking along these lines, starting with simple events—those only talking about the first n bits—and defining simple rules how to construct more complicated events, you will arrive at the formal definition of a probability space in the infinite case (the formal term is sigma-algebra, or σ-algebra; look it up if you want to learn more).

3.1 When Does the First 1 Appear?

Now that we have (informally) introduced how to talk about probabilities on infinite spaces, let us put it to use. Again, consider the space $S = \{0,1\}^\infty$ above. Let T_1 be the time when the first 1 appears. We want to know $\mathbb{E}[T_1]$. In words: on average, how often do we have to toss a coin until we see heads for the first time? There are several ways to compute $\mathbb{E}[T_1]$, all rather instructive. Let us actually consider a more general setting in which the coin is biased and every toss comes up 1 with probability p. In this space, the event "x starts with 1001" has the probability $p \cdot (1-p) \cdot (1-p) \cdot p$. We observe that

$$\Pr[T_1 = n] = \Pr[x_1 = x_2 = \cdots = x_{n-1} = 0, x_n = 1] = p \cdot (1-p)^{n-1} \ ,$$

and therefore

$$\mathbb{E}[T_1] = \sum_{n \geq 1} n \cdot \Pr[T_1 = n] = \sum_{n \geq 1} np(1-p)^{n-1} \ .$$

Lemma 8. $\mathbb{E}[T_1] = 1/p$.

I once had an argument with a (non-mathematician) friend who claimed that this does not need a proof because it is obvious; he thought it was the very definition of probability that this would hold. He was wrong, of course... We will give three different ways of proving this lemma.

Proof 1, Using Calculus. Define $F(p) := \sum_{n\geq 0}(1-p)^n$. This is a geometric series, so we easily compute $F(p) = \frac{1}{p}$. Let us compute $F'(p)$. On the one hand, this is

$$F'(p) = \left(\frac{1}{p}\right)' = \frac{-1}{p^2} \ .$$

Of course, this would require a proof, too, but I assume we have all seen such a proof. One the other hand, differentiating the sum gives

$$F'(p) = \left(\sum_{n\geq 0}(1-p)^n\right) = -\sum_{n\geq 0} n(1-p)^{n-1} = -\sum_{n\geq 1} n(1-p)^{n-1}$$

So $\sum_{n\geq 1} n(1-p)^{n-1} = \frac{1}{p^2}$ and

$$\sum_{n\geq 1} np(1-p)^{n-1} = p\cdot\frac{1}{p^2} = \frac{1}{p} \ .$$

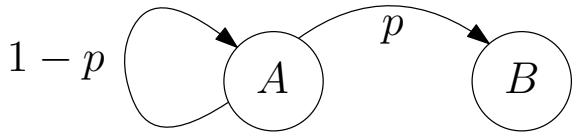
□

Proof 2, by shifting the sum.

$$\begin{aligned}
\mathbb{E}[T_1] &= \sum_{n=1}^{\infty} np(1-p)^{n-1} = \sum_{k=0}^{\infty}(k+1)p(1-p)^k \\
&= (1-p)\sum_{k=0}^{\infty} kp(1-p)^{k-1} + \sum_{k=0}^{\infty} p(1-p)^k \\
&= (1-p)\sum_{k=1}^{\infty} kp(1-p)^{k-1} + \sum_{n=1}^{\infty} p(1-p)^{n-1} \\
&= (1-p)\mathbb{E}[T_1] + \sum_{n=1}^{\infty}\Pr[T_1 = n] \\
&= (1-p)\mathbb{E}[T_1] + 1 \ ,
\end{aligned}$$

from which it follows that $\mathbb{E}[T_1] = 1/p$. It remains to justify the final equality: isn't $\sum_{n=1}^{\infty}\Pr[T_1 = n]$ obviously 1? After all, T_1 has to be *some* natural number. Well, it might be that we *never* see a 1, and in that case T_1 would be infinite, or undefined. Thus, $\sum_{n=1}^{\infty}\Pr[T_1 = n]$ equals 1 minus the probability that we toss only zeroes; the latter probability is 0, as we have already seen. □

Proof 3, by drawing a picture. Consider the following small automaton:



We start in state A. We toss a coin; if we toss a 0 (with probability $1-p$) we stay in state A; from there, it will take us, on expectation, $\mathbb{E}[T_1]$ more coin tosses. If we toss a 1 (probability p) we enter state B and stop. Thus, we can write the following equation:

$$\mathbb{E}[T_1] = 1 + (1-p)\mathbb{E}[T_1] + p \cdot 0 \ .$$

The 1 appears since we have to toss at least one coin. Then, with probability $1-p$, we stay at A, thus have to "start over"; with probability p we move to B, and no further tosses are required. Solving the above equation immediately yields $\mathbb{E}[T_1] = 1/p$. □

In general, for some finite bit string $z \in \{0,1\}^*$, let T_z denote the number of tosses until z appears the first time. For example, if we toss 0010110, then $T_{10} = 4$ and $T_{110} = 7$.

Exercise 3.1. Consider an unbiased coin, i.e., 0 and 1 come up with probability $p = 1/2$ each. Compute $\mathbb{E}[T_{11}]$ and $\mathbb{E}[T_{10}]$. You can choose any of the three proof methods above (but two of them won't be fun).

You might have noticed that $\mathbb{E}[T_{10}] < \mathbb{E}[T_{11]}$, i.e., 10 appears earlier than 11, on average.

Exercise 3.2. Let $\mathcal{E}$ be the event that 10 appears earlier than 11. What is $\Pr[\mathcal{E}]$?

Exercise 3.3. Toss two coins repeatedly, producing two sequences $x_1x_2\ldots$ and $y_1y_2\ldots$. We stop once we see 10 in the first sequence or 11 in the second sequence. Formally, we toss the two coins

$$T := \min(T_{10}(x), T_{11}(y))$$

times.

1. What is $\mathbb{E}[T]$?

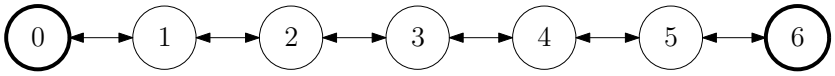
2. What are the probabilities that 10 appears in x (a) before 11 appears in y, (b) at the same time as 11, (c) later than 11?

Hint. I think the only way to solve this exercise without going crazy is by applying proof method 3 from above, drawing a small automaton; or in this case, not so extremely small anymore.

4 Random Walks

4.1 Bounded Random Walks

Imagine an ant walking on this path of length k, in this case $k = 6$:



The ant starts at state j; in every step, it randomly goes left or right, with probability 1/2 each. State 0 and k are goal states, let's imagine 0 is the nest and k is some food source. We can ask the following two questions:

What is the probability that the ant eventually reaches a goal?

What is the expected number of steps until it reaches a goal?

To formalize things, it helps to imagine an infinite string $\mathbf{x} = (x_1, x_2, \dots) \in \{-1, +1\}^\infty$, where -1 means "go left" and $+1$ means "go right". The string is sampled uniformly at random, that is, all bits are independent, and each bit x_i is uniform over $\{-1, +1\}$. It also helps to imagine that the ant continues infinitely, possible beyond below 0 and above k, just that we lose interest once it has reached 0 or k.

Lemma 9. *The probability that the ant, starting at state $j \in \{0, \dots, k\}$, eventually reaches a goal is* 1.

We will give two proofs. A simple one and a more complicated one. We need the more complicated one because we will need parts of it later.

Proof 1. For $j \in \{0, 1, \dots, k\}$, let r_j denote the probability that an ant that starts at state j eventually reach a goal. Obviously $r_0 = r_k = 1$, since the ant starts at a goal in this case. Suppose that not all r_j are 1. Define $r_{\min} = \min(r_0, r_1, \dots, r_k)$ and let j be the smallest index for which $r_j = r_{\min}$. Obviously $r_j < 1$ and $j \in \{1, \dots, k-1\}$ (since $r_0 = r_k = 1$). Starting at j,

the ant randomly moves to $j-1$ or $j+1$ with equal probability. From this it follows that

$$r_j = \frac{r_{j-1} + r_{j+1}}{2} \ .$$

remember that $j \geq 1$ and j is the smallest index for which $r_j = r_{\min}$. This means that $r_{j-1} > r_j$ and $r_{j+1} \geq r_j$. Thus,

$$\frac{r_{j-1} + r_{j+1}}{2} < \frac{r_j + r_j}{2} = r_j \ ,$$

a contradiction. Thus, no r_j can be smaller than 1. □

Proof 2. Define the event $\mathcal{E}$ that the ant never reaches 0 or k. We want to show that $\Pr[\mathcal{E}] = 0$. Let $\mathcal{E}_n$ be the event that the ant does not reach 0 or k within the first n steps. Clearly, $\mathcal{E} = \bigcap_{n \geq 0} \mathcal{E}_n$ and thus $\Pr[\mathcal{E}] \leq \Pr[\mathcal{E}_n]$, for all n. We will bound $\Pr[\mathcal{E}_n]$ from above. Note that if the ant moves right k *times in a row*, then surely she must have reached or even crossed k at some point. Thus, if $\mathcal{E}_n$ happens then the string $x_1 x_2 \dots x_n$ does not contain k consecutive $+1$'s. Similarly, if $\mathcal{E}_{nk}$ happens, then the string $x_1 \dots x_{nk}$ contains no k consecutive $+1$'s. In particular, if we partition $x_1 \dots x_{nk}$ into n blocks of length k each, then no block is completely $+1$. This is good, since different blocks are chosen independently. Moreover, the probability that the ith block is not all $+1$ is exactly $\frac{2^k-1}{2^k}$. Thus,

$$\begin{aligned} \Pr[\mathcal{E}] &\leq \Pr[\mathcal{E}_n] \\ &\leq \Pr[\text{ none of the } n \text{ blocks is completely } +1] \\ &= \left(\frac{2^{k-1}}{2^k}\right)^n \ . \end{aligned}$$

The above inequality holds for all n. Which non-negative real number is at most $\left(\frac{2^{k-1}}{2^k}\right)^n$ for all n? Only this: 0. □

Alright, so now it's settled that the ant ultimately reaches 0 or k. But after how many steps? Let T_j denote the number of steps the ant needs until it reaches 0 or k when starting at j. This is a random variable. If you insist of a formal definition in terms of $\mathbf{x} \in \{-1, +1\}^\infty$, here it is:

$$T_j(\mathbf{x}) = \begin{cases} \min\{T \mid j + x_1 + \dots + x_T \in \{0, k\} & \text{if such a } T \text{ exists} \\ \infty & \text{else.} \end{cases}$$

Yes, T_j can be infinite! The ant could walk from state 2 to 3 and back forever; this has probability 0 but it is not impossible. Gladly, we know that $\Pr[T = \infty] = 0$ since we have just proved it.

Lemma 10. *Starting at $j \in \{0, \ldots, k\}$, the ant reaches one of 0 and k after an expected number of $j(k-j)$ steps. That is, $\mathbb{E}[T_j] = j(k-j)$.*

Proof. Let $t_j := \mathbb{E}[T_j]$. The numbers $t_0, \ldots, t_k$ satisfy the following system of equations:

$$\begin{aligned} t_0 &= 0 \ , \\ t_j &= 1 + \frac{t_{j-1} + t_{j+1}}{2} \quad \text{for } j \in \{1, \ldots, k-1\} \ , \\ t_k &= 0 \ . \end{aligned}$$

It is straightforward to check that $j(k-j)$ is a solution. Is it the only one? Suppose a solution $(s_0, \ldots, s_k)$ and $(t_0, \ldots, t_k)$ are solutions to the above system. Define $\delta_j = s_j - t_j$. Then $\delta_0 = \delta_k = 0$ and $\delta_j = \frac{\delta_{j-1}+\delta_{j+1}}{2}$ for all $1 \leq j \leq k-1$. An argument similar to the one in Proof 1 of Lemma 9 shows that $\delta_j = 0$ for all j, and thus the system has a unique solution.

Are we done? Seems so. But no. Actually, I wrote "the numbers $t_0, \ldots, t_k$". Can we be sure that they are numbers? What??? Look at the definition: $t_j := \mathbb{E}[T_j]$. An expectation can be infinite, in which case it is *not* a number. Thus, to finish the proof, we have to make sure that indeed each expectation $\mathbb{E}[T_j]$ is finite. For this, it helps to write down the definition of the expectation:

$$\begin{aligned} \mathbb{E}[T_j] &= \sum_{n=1}^{\infty} n \cdot \Pr[\text{goal reached after exactly } n \text{ steps}] \\ &\leq \sum_{n=0}^{\infty} (n+1) \cdot \Pr[\text{ the first } n \text{ steps do not reach 0 or } k] \ . \end{aligned}$$

We can now again apply our argument about $\lfloor n/k \rfloor$ blocks of size k each to conclude that the above sum is at most

$$\sum_{n=0}^{\infty} (n+1) \cdot \left(\frac{2^k - 1}{2k} \right)^{\lfloor n/k \rfloor} \ .$$

The above sum clearly converges, for all values of k. Thus, the expectation is finite and we can be confident that each t_j is indeed a real number. □

To summarize, if we start at some $j \in \{0, \ldots, k\}$, then with probability one, the random eventually reaches 0 or k, and it does so in an expected number of $j\,(k-j)$ steps. How likely is it that it reaches k first?

Lemma 11. *Let $0 \leq j \leq k$ be integers. Then the probability that a random walk, starting at j, reaches k before it reaches 0 is $\frac{j}{k}$.*

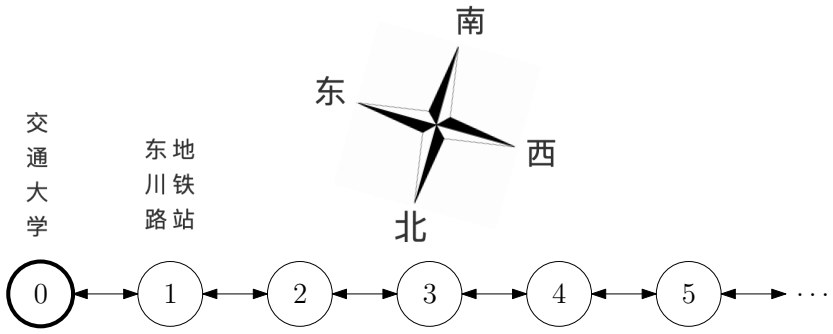
Proof. Let p_j be the probability that the walk starting at j reaches k before 0. We know that $p_0 = 0$ and $p_k = 1$. For $1 \leq j \leq k-1$ we have

$$p_j = \frac{p_{j-1} + p_{j+1}}{2} \ .$$

So we have $k+1$ variables and $k+1$ equations. It is straightforward to check that $p_j = j/k$ is a solution. By the same argument as in the first proof of Lemma 9, one can show that there is a unique solution. □

4.2 Unbounded Random Walks

An incoming undergraduate student arrives at Dongchuan Road subway station and wants to walk to his (yes, it's a male student) new university, Shanghai Jiaotong University. He remembers that he has to walk 1 kilometer along Dongchuan Road. Unfortunately, he does not not in which direction (east would be correct), so he randomly west or east, with probability 1/2 each. Even worse, after walking one kilometer, he gets confused, forgets where he is, and just remembers he has to walk one kilometer along Dongchuan Road. Since Dongchuan Road is infinitely long, it is by no means clear whether he will eventually arrive at the entrance gate of Jiaotong University. We can model the student's behavior as a random walk on $\mathbb{Z}$:

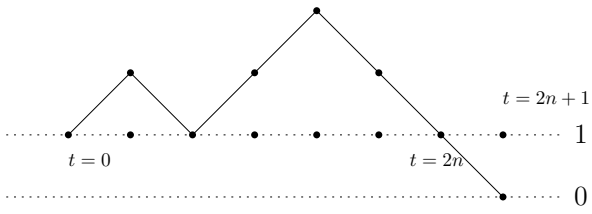


We can ask the following questions:

What is the probability of reaching 0 when starting at 1?
What is the probability of reaching 0 when starting at j?
What is the expected number of steps when starting at 1?
What is the expected number of steps when starting at j?

This scenario is commonly known as *gambler's ruin*. The idea is that a gambler enters a casino with j dollars. In every round, he (yes, it's a male gambler) bets \$1. The casino is *fair*, meaning that with probability $1/2$ he loses \$1 and with probability $1/2$ he wins \$1. He always bets exactly \$1 and leaves the casino once he is "ruined", i.e., once he has lost all his money. Note that "losing all your money" in the scenario corresponds to "successfully arriving at Jiaotong University".

Let us write $\Pr[j \rightsquigarrow i]$ to denote the probability that the student, when starting at state j, will eventually reach state i. It is clear that $\Pr[j \rightsquigarrow j] = 1$, but not much else is clear. To study $\Pr[1 \rightsquigarrow 0]$, let us define the event $\mathcal{E}_n$ that the student reaches 0 after $2n+1$ steps but not earlier. Note that the choice $2n+1$ makes sense: the student must make a positive, odd number of steps. In particular, he must make exactly n steps to the right and $n+1$ steps to the left, and the first $2n$ steps must not lead him to 0. How many possible ways are there to reach 0 in $2n+1$ steps but not before? Recall that this is exactly the number of "mountain ranges", namely C_n, the nth Catalan number:



One of C_n many (in this caes 5) ways to reach 0 in $2n+1$ steps but not earlier.

This gives us an exact formula: $\Pr[\mathcal{E}_n] = C_n \cdot \left(\frac{1}{2}\right)^{2n+1}$ and

$$\Pr[1 \rightsquigarrow 0] = \sum_{n=0}^{\infty} \Pr[\mathcal{E}_n] = \sum_{n=0}^{\infty} \frac{C_n}{2^{2n+1}} = \sum_{n=0}^{\infty} \frac{\binom{2n}{n}}{(n+1)2^{2n+1}} \ .$$

Okay, this is nice, but I have absolutely no idea how to continue from here, i.e., how to compute a closed form for this infinite sum.

Let us try a different approach. Define $r_j := \Pr[j \rightsquigarrow 0]$ and $r := r_1$ for simplicity. To reach 0 when starting at $j \geq 0$, we first have to reach $j-1$; the probability that we ever reach $j-1$ is $\Pr[j \rightsquigarrow j-1]$. It should be clear that this is equal to $\Pr[1 \rightsquigarrow 0] = r$. Once we have reached $j-1$, we must

make our way from $j-1$ to 0; the probability that we ever arrive at 0 is r_{j-1}. Thus, we have just shown that

$$r_j = r \cdot r_{j-1} \ . \tag{2}$$

From this, it follows that $r_j = r^j$. This is already interesting: the probabilities of reaching 0 decay exponentially with the starting point j. But how do we determine r? When starting at 1, the first step will lead us to 0 or 2 with equal probability. Thus,

$$r = \frac{r_0 + r_2}{2} = \frac{1 + r^2}{2} \ . \tag{3}$$

It is straightforward to show that 1 is the unique solution to this quadratic equation. Thus, we have just proved:

Theorem 12. *No matter at which state j he starts, the student eventually reaches 0 with probability 1.*

Next, we can ask how long the student takes on average to reach 0. Let T denote the number of steps until he reaches 0. Note that T can be infinity (as in the previous section), but the probability of this is 0. The following theorem is quite surprising.

Theorem 13. *The expected number of steps from 1 to 0 is infinity.*

That is, although the student almost surely reaches 0, his expected number of steps will be infinite. We give two proofs.

Proof 1. Recall our definition of $\mathcal{E}_n$, the event that the student reaches 0 after $2n+1$ steps but not before. We can write $\mathbb{E}[T]$ as follows:

$$\begin{aligned} \mathbb{E}[T] &= \sum_{n=0}^{\infty} (2n+1) \cdot \Pr[\mathcal{E}_n] \\ &= \sum_{n=0}^{\infty} \frac{(2n+1) \cdot \binom{2n}{n}}{(n+1)2^{2n+1}} \ . \end{aligned}$$

It is straightforward to prove that $\binom{2n}{n} \geq \frac{4^n}{2n+1}$ (try it!), and therefore the above sum is at least

$$\sum_{n=0}^{\infty} \frac{4^n}{(n+1)2^{2n+1}} = \frac{1}{2} \sum_{n=0}^{\infty} \frac{1}{n+1} \ .$$

This is the Harmonic series, and it is well-known that it diverges. Thus, the expectation is infinite. □

Proof 2. Define s_d to be the expected number of steps from 2^d to 0. Thus $\mathbb{E}[T] = s_0$. When starting from 2^d, we will surely reach 0 or 2^{d+1} eventually (see section on bounded random walks); the probability that we reach 2^{d+1} first is 1/2 by symmetry. Without bothering how long this takes on expectation, we observe that we need at least 2^d steps until we reach 0 or 2^{d+1}. Thus,

$$s_d \geq 2^d + \frac{s_{d+1}}{2} \ .$$

A quick induction shows that $s_0 \geq d + s_d$ for all d. Since $s_d \geq 0$ (of course it is non-negative) it means that $s_0 \geq d$ for all d. This is absurd. No number can be larger than every d. We conclude that s_0 is not a number. It is infinite. □

4.3 Biased Random Walks on $\mathbb{Z}$

In the previous section, the random walk passes from j to $j-1$ or $j+1$, each with probability 1/2. For $p \in [0,1]$, we can define the *p-biased random walk*: when in state j, it passes to state $j-1$ with probability p and to $j+1$ with probability $1-p$. Note that for $p = 1/2$, we get the previous *unbiased* random walk. Again, we ask two questions: when starting at j (and of particular interest, at $j = 1$), what is the probability that the p-biased random walk eventually reaches 0? What is the expected number of steps it takes?

Theorem 14. *If $p \leq 1/2$ then the p-biased random walk starting at 1 reaches 0 with probability $r = r(p) = \left(\frac{p}{1-p}\right)$. When starting at j, it reaches 0 with probability $r^j = \left(\frac{p}{1-p}\right)^j$. If $p \geq 1/2$ then the random walk reaches 0 with probability 1.*

Proof. We ask the reader to verify that equation (2) holds for general $p \in [0,1]$, too. From this, we can derive a generalization of (3):

$$r = p + (1-p)r^2 \ .$$

This quadratic equation has two solutions, $r = 1$ and $r = \frac{p}{1-p}$. For $p = 1/2$ those two numbers agree, so $r = 1$ must be the probability of reaching 0. For $p > 1/2$ the second solution is *larger* than 1. Since the probability of reaching 0 is a *probability*, it cannot be larger than 1, so the "true" solution must be $r = 1$.

For $p < 1/2$, it is not clear whether r is 1 or $\frac{p}{1-p}$. We resort to our explicit computation of $\Pr[\mathcal{E}_n]$, the probability that we reach 0 in $2n+1$ steps but not earlier, when starting at 1. For general p, we observe that $\Pr[\mathcal{E}_n] = C_n p^{n+1}(1-p)^n$. Indeed, there are C_n ways to walk from 1 to 0 in $2n+1$ steps without reaching 0 within the first $2n$ steps; each such way makes n steps to the right and $n+1$ steps to the left and thus has probability $p^{n+1}(1-p)^n$. Thus,

$$\Pr[1 \rightsquigarrow 0] = \sum_{n\geq 0} \Pr[\mathcal{E}_n] = \sum_{n\geq 0} C_n p^{n+1}(1-p)^n = p \sum_{n\geq 0} C_n (p(1-p))^n \ . \quad (4)$$

Note that if $p < 1/2$ then $x := p(1-p) < \frac{1}{2} \cdot (1 - \frac{1}{2})$. That is, every term of the sum is strictly smaller than if p were $1/2$, and thus

$$\Pr[1 \rightsquigarrow 0] < \sum_{n\geq C} \frac{C_n}{2^{2n+1}} \ .$$

The latter sum is the probability that the *unbiased* random walk reaches 0, and we have already showed that this is 1. Thus, the probability we are interested in, that the p-biased random walk reaches 0, for $p < 1/2$, is *less than* 1. Thus, $r = 1$ cannot be the true solution, so it must be $\frac{p}{1-p}$. □

5 Generating Functions

5.1 Fibonacci Numbers

In this section, we show the basics and some applications of *generating functions*. Recall the Fibonacci numbers $0, 1, 1, 2, 3, 5, 8, 13, \ldots$ defined by

$$\begin{aligned} F_0 &= 0 \\ F_1 &= 1 \\ F_n &= F_{n-1} + F_{n-2} \quad \text{if } n \geq 2 \ . \end{aligned} \quad (5)$$

In the lecture we have derived an explicit formula for F_n and have seen several combinatorial identities involving the Fibonacci numbers. Let us introduce the concept of generating functions at the example of Fibonacci numbers. We define a function $f(x)$ via the following power series:

$$f(x) := \sum_{n=0}^{\infty} F_n x^n \ . \quad (6)$$

This function is defined for those $x \in \mathbb{R}$ for which the above power series converges. Note that we can do such a thing with every sequence of numbers. What is the one thing we know about the Fibonacci numbers? Exactly: the recurrence (5). Let us plug (5) into (6).

$$\begin{aligned}
f(x) &= F_0 + xF_1 + \sum_{n=2}^{\infty} F_n x^n \\
&= x + \sum_{n=2}^{\infty} (F_{n-1} + F_{n-2}) x^n \\
&= x + \sum_{n=2}^{\infty} F_{n-1} x^n + \sum_{n=2}^{\infty} F_{n-2} x^n \\
&= x + \sum_{i=1}^{\infty} F_i x^{i+1} + \sum_{j=0}^{\infty} F_j x^{j+2} \qquad \text{(via } i := n-1 \text{ and } j := n-2\text{)} \\
&= x + x \sum_{i=0}^{\infty} F_i x^i + x^2 \sum_{j=0}^{\infty} F_j x^j \qquad \text{(since } F_0 = 0\text{)} \\
&= x + xf(x) + x^2 f(x) \ .
\end{aligned}$$

Thus, we have just shown that $f(x) = x + xf(x) + x^2 f(x)$. Solving for $f(x)$, we see that

$$f(x) = \frac{x}{1 - x - x^2} \ .$$

This is somewhat amazing: we have not yet derived an explicit formula for F_n; well, actually, we did in the lecture, but it was not super easy; however, finding a closed form for the generating function (6) was quick and painless.

Great, but what new insights do we gain by this? Let me show you two applications. First, note that if we write $y = x + x^2$ then $f(x) = x \cdot \frac{1}{1-y}$. The term $\frac{1}{1-y}$ can be written as a geometric series $\sum_{n \geq 0} y^n$. Thus,

$$\begin{aligned}
f(x) = x \cdot \frac{1}{1 - (x + x^2)} &= x \cdot \sum_{n=0}^{\infty} (x + x^2)^n \\
&= \sum_{n=0}^{\infty} x^{n+1} (1 + x)^n \\
&= \sum_{n=0}^{\infty} x^{n+1} \sum_{k=0}^{n} \binom{n}{k} x^k
\end{aligned}$$

Let us switch the order of summation; to facilitate things, observe that $\binom{n}{k} = 0$ once $k > n$, so we can simply sum over all $k \geq 0$ and write $\sum_k$ instead of $\sum_{k=0}^{\infty}$. All sums are understood to be over all non-negative integers, unless specified otherwise. Also, we use the notation $[P]$ where P is a statement and $[P]$, is 1 if P is true and 0 if P is false.

$$\begin{aligned} f(x) &= \sum_{n,k} x^{n+k+1} \binom{n}{k} \\ &= \sum_{n,k,m} [m = n+k] x^{m+1} \binom{n}{k} \\ &= \sum_{n,k,m} [n = m-k] x^{m+1} \binom{m-k}{k} \\ &= \sum_m x^{m+1} \sum_k \binom{m-k}{k} . \end{aligned}$$

Okay, this seems complicated, but once you get used to this kind of sum manipulation, you can do it in one line and just say "by switching the order of summation we obtain...". Note that we now have *two* power series for $f(x)$. Namely,

$$\sum_n F_n x^n = f(x) = \sum_n x^{n+1} \sum_k \binom{n-k}{k} .$$

This identity holds for all x for which the sum converges; without formally proving it, let me claim that the two power series must agree in every coefficient. That is, the coefficient of x^n in the left power series (which is F_n) must be the same as in the right (which is $\sum_k \binom{n-1-k}{k}$. Let us try for $n = 8$ and $n = 9$:

$$\begin{aligned} \binom{7}{0} + \binom{6}{1} + \binom{5}{2} + \binom{4}{3} &= 1 + 6 + 10 + 4 = 21 = F_8 \\ \binom{8}{0} + \binom{7}{1} + \binom{6}{2} + \binom{5}{3} + \binom{4}{4} &= 1 + 7 + 15 + 10 + 1 = 34 = F_9 . \end{aligned}$$

Let us summarize what we have just proved:

Lemma 15. $F_n = \sum_k \binom{n-1-k}{k}$.

Sure, this result can be proven by other means, too. I guess by induction on n (I have not tried this) or by purely combinatorial reasoning (I have done

this).

We have seen that finding a closed form for $f(x) := \sum_n F_n x^n$ helps us discover new identities. It can also helps us come up with an exact formula for F_n. For this, we want to write $\frac{x}{1-x-x^2}$ in partial fractions. After solving $1 - x - x^2$ and doing some math, we find that we can write $1 - x - x^2 = (1 - x\phi)(1 - x\psi)$ for $\phi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$ (this is easy to check). To expand $\frac{x}{1-x-x^2}$ in partial fractions, we write

$$f(x) = \frac{x}{1 - x - x^2} = \frac{x}{(1 - x\phi)(1 - x\psi)}$$

$$\frac{a}{1 - x\phi} + \frac{b}{1 - x\psi} = \frac{a(1 - x\psi) + b(1 - x\phi)}{(1 - x\phi)(1 - x\psi)}$$

For the two expressions to be the same, we want $a(1 - x\psi) + b(1 - x\phi) = x$. This holds for $a = \frac{1}{\phi-\psi} = \frac{1}{\sqrt{5}}$ and $b = -a$. Thus, we obtain that

$$f(x) = \frac{1}{\sqrt{5}} \cdot \left(\frac{1}{1 - x\phi} + \frac{1}{1 - x\psi} \right) .$$

By applying the formula for geometric series backwards, we see that this is

$$f(x) = \frac{1}{\sqrt{5}} \cdot \left(\sum_{n \geq 0} \phi^n x^n - \sum_{n \geq 0} \psi^n x^n \right) .$$

Again, we have written $f(x)$ as a power series. What is the coefficient of x^n in the series above? Surely it is $\frac{1}{\sqrt{5}}(\phi^n - \psi^n)$; but it is also F_n. We have just derived that

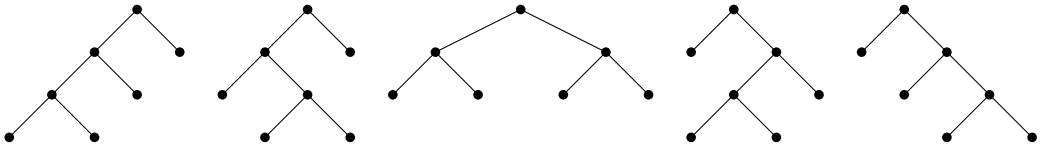
$$F_n = \frac{1}{\sqrt{5}} \left(\phi^n - \psi^n \right)$$

for $\phi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$. Alright, we have derived this formula before, by computing the eigenvectors and eigenvalues of the matrix $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. Is the above method using generating functions easier? Might depend on your taste. But it is certainly good to have a new tool in the box.

Let us pause for a minute. Why was it to easy to find a closed form for $f(x)$? Certainly because the Fibonacci numbers F_n follow a simple recurrence. In the next section we will see how to use a more complicated recurrence to find a generating function.

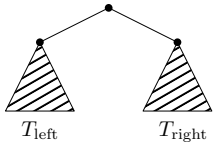
5.2 The Catalan Numbers

Remember the Catalan numbers C_n? They can be defined in many ways; for now, let us define C_n as the numbers of complete binary trees with n inner nodes. Recall that in a complete binary tree, every node has either two children (then it's called an inner node) or no children (then it is a leaf).



There are five complete binary trees with 3 inner nodes: $C_3 = 5$.

For small n, we find that $C_0 = 1$ (a single vertex), $C_1 = 1$ (a root with two children), $C_2 = 2$ (draw it yourself), $C_3 = 5$ (see above). Drawing all C_4 binary trees with 4 inner nodes would take too much time and space, so let's try to find a formula. Surely, if $n \geq 1$, then the root has two children, and we can decompose the tree T into two subtrees T_{left} and T_{right}:



If the T_{left} has i inner nodes then T_{right} has $n - i - 1$ inner nodes, and there are $C_i \cdot C_{n-1-i}$ such trees. The value of i ranges from 0 to $n - 1$, and thus

$$C_n = \sum_{i=0}^{n-1} C_i C_{n-i-1} \quad \text{if } n \geq 1 \ , \tag{7}$$

is our sought recurrence. Let's see how this helps us find a closed form for the generating function:

$$\begin{aligned}
c(x) &:= \sum_{n\geq 0} C_n x^n = 1 + \sum_{n\geq 1} C_n x^n \\
&= 1 + \sum_{n\geq 1} \sum_{i=0}^{n-i-1} C_i C_{n-i-1} x^n \\
&= 1 + \sum_{n\geq 1} \sum_{i=0}^{n-1} C_i x^i C_{n-i-1} x^{n-i-1} x \\
&= 1 + x \sum_{n\geq 1} \sum_{i} \sum_{j} [j = n-i-1] C_i x^i C_j x^j \\
&= 1 + x \sum_{i} C_i x^i \sum_{j} C_j x^j \sum_{n\geq 1} [i + j = n-1]
\end{aligned}$$

Observe that the condition $i + j = n - 1$ is satisfied for exactly one value of n, namely $n = i + j + 1$, thus the inner sum always evaluates to 1. Thus, the above sum simplifies to

$$c(x) = 1 + x \sum_{i} C_i x^i \sum_{j} C_j x^j = 1 + x c(x) c(x) \ .$$

For every fixed x, we get a quadratic equation for $c = c(x)$, namely $c = 1 + xc^2$ or $xc^2 - c + 1 = 0$. This has two solutions:

$$c_{1,2} = \frac{1 \pm \sqrt{1 - 4x}}{2x} \ .$$

So we get two solutions; both are well-defined for all $0 < x \leq 1/2$, but which one is $c(x)$? Taking $\lim_{x\to 0}$ we see that

$$\begin{aligned}
\lim_{x\to 0} c_1(x) &= \lim_{x\to 0} \frac{1 + \sqrt{1-4x}}{2x} = \infty \text{ and} \\
\lim_{x\to 0} c_2(x) &= \lim_{x\to 0} \frac{1 - \sqrt{1-4x}}{2x} \\
&= \frac{\lim_{x\to 0}(1 - \sqrt{1-4x})'}{\lim_{x\to 0}(2x)'} \qquad \text{(l'Hopital)} \\
&= \frac{\lim_{x\to 0} \frac{2}{\sqrt{1-4x}}}{2} = \frac{2}{2} = 1 \ , \quad \text{whereas} \\
\lim_{x\to 0} c(x) &= \lim_{x\to 0} \sum_{n\geq 0} C_n x^n = C_0 = 1 \ .
\end{aligned}$$

Thus, c_2 is the correct solution, and

$$c(x) = \sum_{n \geq 0} C_n x^n = \frac{1 - \sqrt{1-4x}}{2x} \ . \tag{8}$$

This equation holds whenever $\sum_{n \geq 0} C_n x^n$ converges. A quick estimate shows that $C_n \leq 2^{2n+1} = 2 \cdot 4^n$ (for example, we can encode a binary tree using $2n+1$ bits by visiting the tree by depth-first search and for each node writing down whether it is an inner node or a leaf). Thus, the sum converges for every $x < 1/4$. In the lecture, we derived an exact formula for C_n, and from this were able to conclude that $c(x)$ converges for $x = 1/4$, too. However, from the above calculation we do not know this yet. We now know that $c(x)$ *diverges* if $x > 1/4$: indeed, if it did converge, then (8) would hold; but this is absurd: the left-hand side would be some real number, but the right-hand side would not be real.

Again, we were able to obtain a closed form for the generating function $c(x) := \sum_{n \geq 0} C_n x^n$ because we had a recurrence for C_n. Let me show you an alternative way of computing $c(x)$ by recycling something we have already seen in Section 4.3. Consider the p-biased random walk on $\mathbb{Z}$ starting at 1, for some $p \leq 1/2$. By Theorem 14 we know that the probability that we eventually reach 0 is $\Pr[1 \rightsquigarrow 0] = \frac{p}{1-p}$. On the other hand, by summing over all walks of length $2n+1$ that start at 1, do n right steps, $n+1$ left steps, and do not reach 0 within the first $2n$ steps, we already saw in (4) that

$$\Pr[1 \rightsquigarrow 0] = \sum_{n \geq 0} C_n p^{n+1} (1-p)^n \ .$$

Combining these two results, we see that for $p \leq 1/2$, it holds that

$$\begin{aligned} \frac{p}{1-p} &= \sum_{n \geq 0} C_n p^{n+1} (1-p)^n = p \cdot \sum_{n \geq 0} C_n \left(p(1-p)\right)^n \\ &= pc(p(1-p)) \ . \end{aligned}$$

Thus, $c(p(1-p)) = \frac{1}{1-p}$. Observe that when p ranges over $[0, 1/2]$, then $p(1-p)$ rises monotonically from 0 to $1/4$. Thus, for every $x \in [0, 1/4]$ there is exactly one $p \in [0, 1/2]$ for which $p(1-p) = x$. Solving the quadratic equation $p(1-p) = x$ for p, we see that

$$p_{1,2} = \frac{1 \pm \sqrt{1-4x}}{2} \ .$$

Of these two solutions, $p_2 = \frac{1-\sqrt{1-4x}}{2}$ is the one in $[0, 1/2]$, and therefore

$$c(x) = c(p_2(1-p_2)) = \frac{1}{1-p_2} = \frac{2}{1+\sqrt{1-4x}} \ .$$

Alright, now we derived a closed formula for $c(x)$ in two different ways. The first one, using the recurrence, gives us $c(x) = \frac{1-\sqrt{1-4x}}{2x}$, the second, using p-biased random walks on $\mathbb{Z}$, gives us $c(x) = \frac{2}{1+\sqrt{1-4x}}$. Luckily, these two expressions are the same, for all $0 < x < 1/4$.

Remember how we computed $\Pr[1 \to 0]$? It was a lot of probabilistic reasoning. If you felt uncomfortable with that, despair not. Here is an alternative way. We can use our (8) first computation of $c(x)$, the one using the recurrence formula for C_n, to compute $\Pr[1 \to 0]$:

$$\begin{aligned}
\Pr[1 \to 0] &= \sum_{n \geq 0} C_n p^{n+1} (1-p)^n = pc(p(1-p)) \\
&= p \cdot \frac{1 - \sqrt{1-4p(1-p)}}{2p(1-p)} \qquad \text{(by (8) with } x = p(1-p)) \\
&= \frac{1 - \sqrt{1-4p+4p^2}}{2(1-p)} = \frac{1 - \sqrt{(1-2p)^2}}{2(1-p)} \\
&= \frac{1 - |1-2p|}{2(1-p)} \ .
\end{aligned}$$

For $p \leq 1/2$, this is $\frac{p}{1-p}$. For $p \geq 1/2$, this is 1.

Alright, so now we walked all the way in both directions: we derived a closed formula for $c(x)$ using our formula for $\Pr[1 \to 0]$, which we obtained by probabilistic reasoning; then we derived a closed formula for $\Pr[1 \to 0]$ without much probabilistic reasoning, but using a closed formula for $c(x)$, which we have computed earlier using the recurrence for C_n. We are exhausted and decide to leave the Catalan numbers alone for the time being.

5.3 Partition Numbers

We played around with generating functions for the Fibonacci numbers and the Catalan numbers and showed how one can use them to derive interesting results. However, we had already derived all of these results via combinatorial means. In this section, I want to show you a result for which you have to use generating functions; or at least you *should* use it, as a proof that avoids

them altogether might become really nasty.

Let me introduce to you the partition numbers P_n. In how many ways can we write the natural number n as a sum of natural numbers? Let us try for $n = 4$: 4, $3+1$, $2+2$, $1+1+1+1$. Note that we do not care about the order of the summands, so writing 4 as $1+2+1$ is the same as writing it as $1+1+2$, which we have already listed. There are five ways to write 4 as a sum of integers.

Definition 16. *The nth partition number, P_n, is the number of ways to write n as a sum of positive integers. More formally, it is the number of finite sequences $(x_1, x_2, \dots x_k) \in \mathbb{N}^*$ such that $x_1 \geq x_2 \geq \cdots \geq x_k \geq 1$ and $x_1 + \cdots + x_k = n$.*

We already have seen that $P_4 = 5$. It is easy to see that $P_3 = 3$, since 3 can be written as $1+1+1$, $1+2$, and 3. Also, $P_2 = 2$ and $P_1 = 1$ is easy. What is P_0? Well, the empty sum, containing no terms, is a sum also, and it evaluates to 0, so $P_0 = 1$. Let us quickly do P_5 and P_6 as well:

$$P_5 : 5, 41, 32, 311, 221, 21111, 111111$$
$$P_6 : 6, 51, 42, 411, 33, 3213111, 222, 2211, 21111, 11111$$

So $P_5 = 7$ and $P_6 = 11$. To summarize, for $n = 2, 3, 4, 5, 6$ we find that $P_n = 2, 3, 5, 7, 11$. Whoa! The prime numbers! This is surprising, since the definition of P_n makes no reference to multiplication! Unfortunately, when going a bit further, we see that $P_7 = 15$, very much not a prime number.

5.3.1 A Two-Dimensional Recurrence

Let us try to find a recurrence for P_n. We define the auxiliary numbers P_n^i, the number of ways to write $n = x_1 + \cdots + x_k$ with the additional condition that $x_1 \leq i$. If we write n as such a sum, note that $x_2 + \cdots + x_k = n - x_1$ and $x_2 \leq x_1$. Thus, there are exactly $P_{n-x_1}^{x_1}$ ways to write n as a sum in which the largest summand is x_1. Summing over all x_1, we obtain the recurrence

$$P_n^i = \sum_{x=1}^{i} P_{n-x}^x \ . \tag{9}$$

I let you figure out the base cases yourself, like P_n^0 and so on. The recurrence in (9) is certainly very useful. For example, we can now easily write a program

that computes the P_n^i—and the P_n—using dynamic programming. It's a bit unsatisfying, though, that (9) is a "two-dimensional" recurrence, using two variables, namely n and i. Also, our dynamic program will have complexity $\Theta(n^2)$. Can we find a "one-dimensional" recurrence that writes P_n as a sum of $P_0, \dots, P_{n-1}$?

5.3.2 Generating Function for P_n

So far, we know very little about the P_n. How can we find a closed form for the generating function $p(n) := \sum_{n\geq 0} P_n x^n$? For this, let us slightly change the way we "encode" a sum like $15 = 4+3+3+2+1+1+1$. For an infinite sequence $\mathbf{a} = (a_1, a_2, \dots) \in \mathbb{N}_0^\infty$ of integers, define its *weight* $w(\mathbf{a}) := \sum_{i=1}^\infty i a_i$. Note that $w(\mathbf{a})$ is finite if and only $a_i = 0$ for all but *finitely many* indices i.

Proposition 17. *P_n is the number of $\mathbf{a}$ with $w(\mathbf{a}) = n$.*

Proof. Let me give you a proof by example. If we write an integer as a sum of positive integers, like $15 = 4+3+3+2+1+1+1$, read this from right to left and group equal numbers together. Thus, the above sum becomes $15 = 3\times 1 + 1\times 2 + 2\times 3 + 1\times 4 + 0\times 5 + 0\times 6 + 0\times 7\dots$ and can be encoded as $\mathbf{a} = (3,1,2,1,0,0,0,\dots)$ with $w(\mathbf{a}) = 15$.

To give a second example, $7 = 6+1$ can written as $1\times 1 + 0\times 2 + 0\times 3 + 0\times 4 + 0\times 5 + 1\times 6 + 0\times 7\dots$ and encoded as $\mathbf{a} = (1,0,0,0,0,1,0,0,\dots)$ with $w(\mathbf{a}) = 7$. □

This new representation helps us find a formula for the generating function:

$$\begin{aligned} p(x) &= \sum_{n\geq 0} P_n x^n = \sum_{n\geq 0} x^n |\{\mathbf{a}\in\mathbb{N}_0^\infty : w(\mathbf{a}) = n\}| \\ &= \sum_{\mathbf{a}\in\mathbb{N}_0^\infty : w(\mathbf{a})\in\mathbb{N}_0} x^{w(\mathbf{a})} \ . \end{aligned}$$

Let us adopt the convention that $x^\infty = 0$ (which kind of makes sense for

$x < 1$). Also note that $x^{w(\mathbf{a})} = x^{\sum_i i a_i} = \prod_i (x^i)^{a_i}$. Then we obtain

$$\begin{aligned}
p(x) &= \sum_{\mathbf{a} \in \mathbb{N}_0^\infty} x^{w(\mathbf{a})} = \sum_{a_1} \sum_{a_2} \sum_{a_3} \dots x^{w(\mathbf{a})} \\
&= \sum_{a_1} \sum_{a_2} \sum_{a_3} \cdots \prod_i \left(x^i\right)^{a_i} \\
&= \sum_{a_1} \left(x^1\right)^{a_i} \times \sum_{a_2} \left(x^2\right)^{a_2} \times \sum_{a_3} \left(x^3\right)^{a_3} \times \dots \\
&= (1 + x + x^2 + x^3 + \dots) \times (1 + x^2 + x^4 + x^6 + \dots) \times (1 + x^3 + x^6 + x^9 + \dots) \times \cdots \\
&= \frac{1}{1-x} \times \frac{1}{1-x^2} \times \frac{1}{1-x^3} \times \cdots
\end{aligned}$$

Okay, I agree this looks like dark magic. We certainly have to justify that it makes sense to sum over all $\mathbf{a} \in \mathbb{N}^\infty$ (since this is an uncountable set, if you know what that means), and we would have to check in detail the definition of this infinite sequence of sums. For the time being, let us ignore all these concerns and just check by hand that if you expand

$$(1 + x + x^2 + x^3 + \dots) \times (1 + x^2 + x^4 + x^6 + \dots) \times (1 + x^3 + x^6 + x^9 + \dots) \times \cdots ,$$

then each term that gives you x^7 really corresponds to a way of writing 7 as a sum of positive integers. As long as we believe that, we have proved that

Lemma 18. *The generating function of the partition numbers P_n is $p(x) := \sum_{n \geq 0} P_n x^n = \frac{1}{1-x} \cdot \frac{1}{1-x^2} \cdot \frac{1}{1-x^3} \cdots$.*

Have a look at your calculus textbook and prove that for every $0 \leq x < 1$, this infinite product converges to some finite real number.

Define

$$q(x) := \frac{1}{p(x)} = (1-x) \cdot (1-x^2) \cdot (1-x^3) \cdots \tag{10}$$

This is the generating function of some sequence Q_n. What are the Q_n? Well, we can find the Q_n by simply expanding this infinite product. Suppose for the moment that we can find a "nice" formula for the Q_n. How would

that be helpful? Well, since $p(x) \cdot q(x) = 1$ by definition, we see that

$$\begin{aligned} 1 = p(x) \cdot q(x) &= \sum_i P^i x^i \sum_j Q^j x^j \\ &= \sum_{i,j,n} [n = i + j] P_i Q_j x^n \\ &= \sum_n x^n \sum_{i,j:i+j=n} P_i Q_j \ . \end{aligned}$$

The right-hand side is a power series. The left-hand side is a power series, too. Namely, it is $1 + 0 \cdot x + 0 \cdot x^2 + 0 \cdot x^3 + \ldots$. Comparing coefficients for each monomial, we observe that $\sum_{i,j:i+j=n} P_i Q_j$ is 1 if $n = 0$ and 0 otherwise. This means that $P_n Q_0 + P_{n-1} Q_1 + \cdots + P_1 Q_{n-1} + P_0 Q_n = 0$. Since $Q_0 = 1$ (check this!), we obtain

$$P_n = -Q_1 P_{n-1} - Q_2 P_{n-2} - \cdots - Q_{n-2} P_2 - Q_{n-1} P_1 - Q_n P_0 \ . \tag{11}$$

This is splendid! It is a one-dimensional recurrence for P_n! Just what we wanted! However, it is only useful if we have a simple way to compute that Q_i. This is what we will do now.

5.3.3 Computing the Q_n

Note that to get a term x^n, we have to pick the $-x^i$ in finitely many parentheses, and in the rest we have to pick 1. This "choice" can be encoded as a finite subset $S \subseteq \mathbb{N}$. To be more formal, write $w(S) := \sum_{i \in S} i$ for a finite subset $S \subseteq \mathbb{N}$. Let R_n be the number of finite sets S with $w(S) = n$. That is, the number of ways to write n as a sum of *distinct positive integers*. Note that

$$(1 + x) \cdot (1 + x^2) \cdot (1 + x^3) \cdots = \sum_{n=0}^{\infty} R_n x^n \ , \tag{12}$$

since each term in the expansion of the product corresponds to a set of natural numbers, e.g., the choice, $1 \cdot x^2 \cdot 1 \cdot x^4 \cdot x^5 \cdot 1 \cdot 1 \cdots$ corresponds to the set $\{2, 4, 5\}$. The monomials multiply to x^{11}, just as $w(\{2, 4, 5\}) = 11$. We have found that (12) is the generating function of the R_n. However, we are not interested in (12) but in (10). Again, to obtain the term x^n in the expansion of (10), we have to select a subset $S \subseteq \mathbb{N}$ with $w(S) = n$. However, if $|S|$ is odd we obtain $-x^n$; if $|S|$ is even we obtain x^n. Thus, we conclude:

Lemma 19. *Q_n is the number of sets $S \subseteq \mathbb{N}$ with $w(S) = n$ and $|S|$ even,* minus *the number of such sets S with $w(S) = n$ and $|S|$ odd.*

From now on, everything will be pure combinatorial reasoning; we don't need generating functions anymore. First, let us get some feeling by doing some examples:

$$\begin{aligned} 8 &= 8 && \text{(odd)} \\ &= 7 + 1 && \text{(even)} \\ &= 6 + 2 && \text{(even)} \\ &= 5 + 3 && \text{(even)} \\ &= 5 + 2 + 1 && \text{(odd)} \\ &= 4 + 3 + 1 && \text{(odd)} \end{aligned}$$

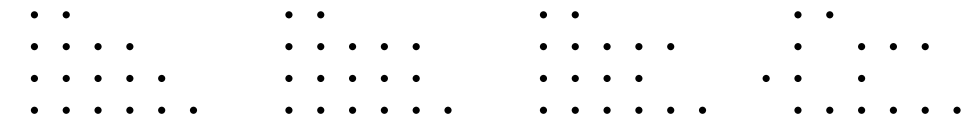
We get three odd sets and three even sets and conclude that $Q_8 = 0$. Coincidence? Let's try $n = 11$:

$$\begin{aligned} 11 &= 11 && \text{(odd)} \\ &= 10 + 1 && \text{(even)} \\ &= 9 + 2 && \text{(even)} \\ &= 8 + 3 && \text{(even)} \\ &= 8 + 2 + 1 && \text{(odd)} \\ &= 7 + 4 && \text{(even)} \\ &= 7 + 3 + 1 && \text{(odd)} \\ &= 6 + 5 && \text{(even)} \\ &= 6 + 4 + 1 && \text{(odd)} \\ &= 6 + 3 + 2 && \text{(odd)} \\ &= 5 + 4 + 2 && \text{(odd)} \\ &= 5 + 3 + 2 + 1 && \text{(even)} \end{aligned}$$

We get twelve different ways, six of them even, six of them odd, so $Q_{11} = 0$. Could it be that $Q_n = 0$ for all n? Certainly not, since $Q_0 = 1$, $Q_1 = -1$, $Q_2 = -1$ are easy enough to check. Playing around even more, we discover that Q_n is indeed 0 for "most" n. Let us try to prove this:

Lemma 20. *$Q_n = 0$ unless $n = ...$ (we don't know yet what to put here).*

Proof. To prove $Q_n = 0$, we have to show that there are equally many even sets S summing up to n as odd sets. One way to show this is to "pair them up", to match every even set S with an odd set S in a unique way. That's what we're going to do. We introduce the notion of a *hill*, a useful way to represent a finite set of natural numbers:



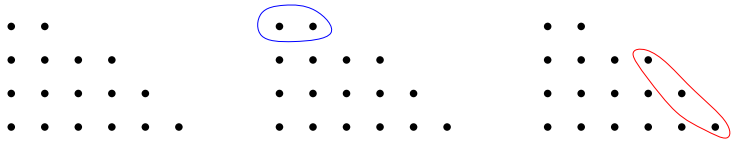
This is a hill, representing the set $\{6, 5, 4, 2\}$.

Not a hill: the same number appears twice.

Not a hill: the order is wrong.

Not a hill, either.

A hill has a *top* and a *front*, as illustrated here.

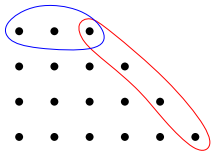


$\{6, 5, 4, 2\}$ as a hill

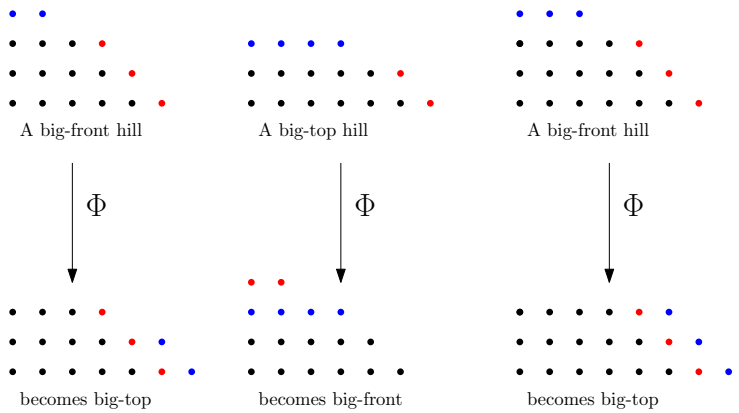
The top of the hill

The front of the hill

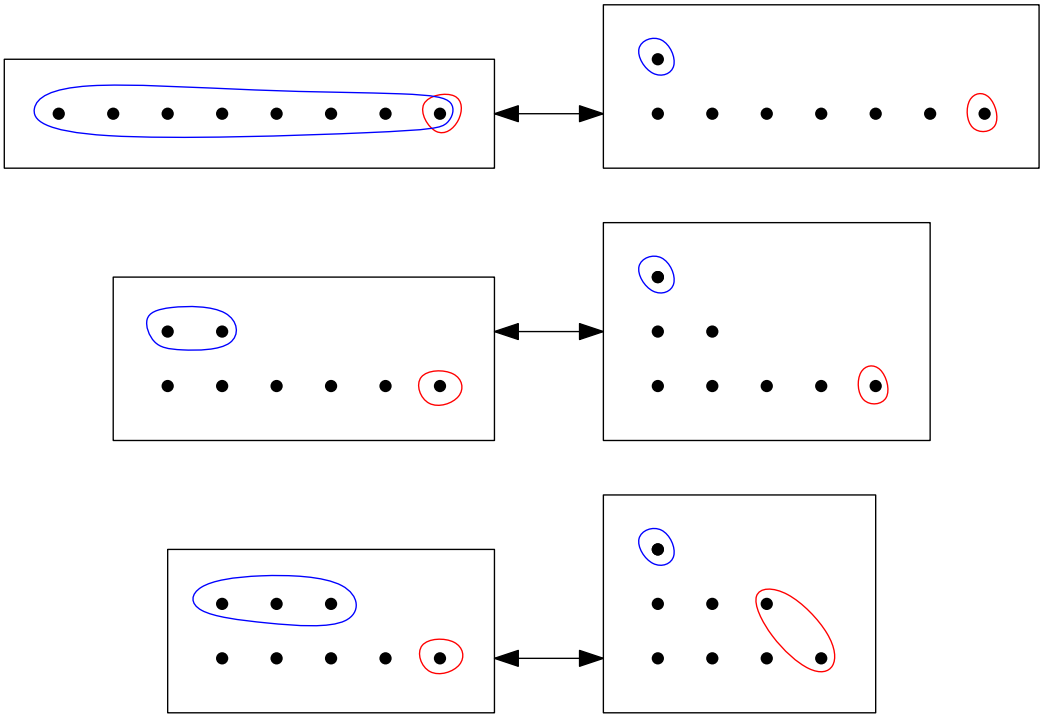
The *top* is easy to understand. The *front* is less intuitive. It contains the right-most point of the bottom row, and then the right most points of the rows above, as long as those rows shrink by one in each step. I leave it to you to come up with a formal definition. Note that top and front might overlap:



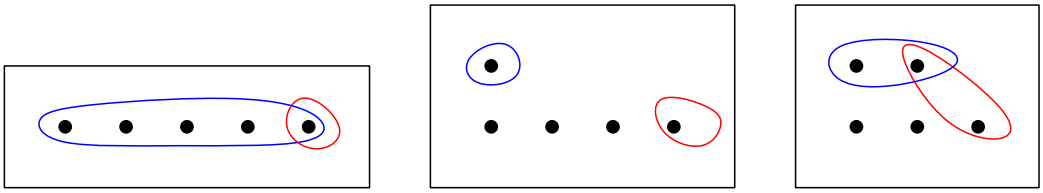
Let f and t denote the size of the front and the top, respectively. We classify hills into two types: if $t > f$, we call it big-top. If $f \geq t$, we call it big-front. We now define an operation Φ on hills: if the hill is big-top, we remove the front and add it at the top; if it is big-front, we remove the top and add it to the front:



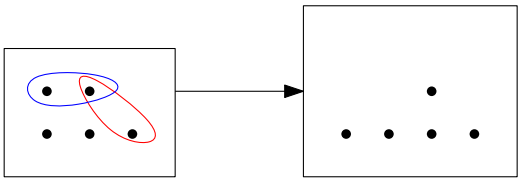
For a hill, let f, t denote the size of the front and top, respectively, and f', t' those sizes after Φ has been applied. If the hill is big-top, i.e., $f < t$ then Φ moves it to the top, so $t' = f$. The size of the new front may be larger or equal: $f' \geq f$, it depends on the specific numbers. In any case, $t' = f < f'$, so the new hill is big-front. If the hill is big-front, i.e., $f \geq t$, then Φ moves t to the front. Thus $f' = t$. The new top will be one level below, thus larger: $t' > t$. Thus the new hill is big-top. Also note that applying Φ twice takes us back to the original hill: $\Phi \circ \Phi$ is the identity function. Finally, recall that a hill represents a set S of natural numbers, where $|S|$ is the height of the hill and $w(S) = \sum_{i \in S} i$ is the number of points in the hill. Φ changes the height by 1 but does not change the number of points. Thus, it matches an even set with an odd set. We conclude that the number of even sets and odd sets are equal, and $Q_n = 0$. The picture below illustrates this matching for $n = 8$.



But wait, this cannot be true: it fails for $n = 0, 1, 2$ and in fact many more. For example, 5 can be written as 5, $4 + 1$, $3 + 2$, thus $Q_5 = 1$, right? What went wrong in the above proof? Well, let's draw the three hills for $n = 3$ and see what goes wrong:



Okay, so Φ would transform the first hill into the second, and the second back into the first. But what happens to the third hill? Let's look into the definition of Φ. The hill $\{3, 2\}$ is big-front, since $f = 2$ and $t = 2$ and thus $f \geq t$. Accordingly, Φ moves the top to the front:



The hill $\{3, 2\}$. $\Phi(\{3, 2\})$ is not a hill.

Thus, the hill $\{3,2\}$ survives the cancellation. It has even height, and thus $Q_5 = 1$. When does this problem arise? Well, first of all, the front and the top have to intersect. This happens only if the hill has height f and is of the form $\{a, a-1, \ldots, a-f+1\}$. This alone might not be a problem. However, we get a problem in two cases:

First, if the hill is big-front, $f \geq t$, we get a problem if the front does not have enough "space" to take in the top, i.e., if $f = t$. Thus, if the hill is of the form

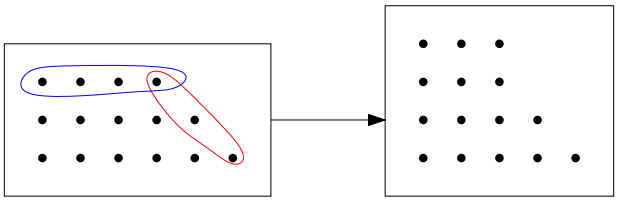
$$\{a, a-1, \ldots, a-f+1\} \text{ with } a-f+1 = f \ . \tag{13}$$

In this case, $a = 2f+1$, and

$$\begin{aligned} n = w(S) &= a + (a-1) + \cdots + (a-f+1) = fa - (1+2+\cdots+f-1) \\ &= af - \binom{f}{2} = f(2f+1) - \frac{f(f-1)}{2} = \frac{4f^2+2f-f^2+f}{2} = \frac{f(3f-1)}{2} \ . \end{aligned}$$

Plugging in $f = 1, 2, \ldots$, this happens for $n = 1, 5, 12, 22, \ldots$

Second, if the hill is big-top, i.e., $t > f$, we move the front to the top. This causes a problem if the old top is too small to "support" the new top. Since top and front intersect, the old top shrinks by 1 and the new top has size f. This is a problem if now the two top rows have equal size, i.e., $t = f+1$:



The hill $\{6,5,4\}$ $\quad$ $\Phi(\{6,5,4\})$ is not a hill

So this happens if $t = f-1$ and thus the set is

$$\{a, a-1, \ldots, a-f+1\} \text{ with } a-f+1 = f+1. \tag{14}$$

In this case, $a = 2f$, and again

$$n = af - \binom{f}{2} = 2f^2 - \frac{f(f-1)}{2} = \frac{3f^2+f}{2} \ .$$

Plugging in $f = 1, 2, \ldots$, this happens for $n = 2, 7, 15, 26$. We can further simplify things by observing that $\frac{3f^2+f}{2} = \frac{3k^2-k}{2} = \frac{k(3k-1)}{2}$ for $k := -f$. This

way, we can tie Case 1 and Case together and see that we get an exception if $n = \frac{k(3k-1)}{2}$ for all $k \in \mathbb{Z} \setminus \{0\}$. Also note that in these exceptional cases, the size of S, i.e., the height of the hill, is f (i.e., k or $-k$), and thus the sign is $(-1)^k$. Finally, for $k = 0$ we already know that $Q_0 = 1$, and thus the formula is correct for $k = 0$, too. □

We can now restate the lemma in its complete form:

Lemma 21. *The coefficients Q_n of x^n in the expansion of $q(x) = (1-x)(1-x^2)(1-x^3)\cdots$ is*

$$Q_n = \begin{cases} (-1)^k & \text{if } n \text{ is of the form } \frac{k(3k-1)}{2} \text{ for some } k \in \mathbb{Z}, \\ 0 & \text{otherwise.} \end{cases}$$

Combining this with (11) above, we obtain

$$\begin{aligned} P_n &= -Q_1 P_{n-1} - Q_2 P_{n-2} - \cdots - Q_{n-1} P_1 - Q_n P_0 \\ &= \sum_{i=1}^{\infty} (-Q_i) P_{n-i} \qquad \text{(with the convention that } P_i = 0 \text{ for } i < 0) \\ &= \sum_{k \in \mathbb{Z}\setminus\{0\}} (-1)^{k+1} Q_{k(3k-1)} \cdot P_{n-k(3k-1)} \ . \end{aligned}$$